ME1103
Principles of Mechanics and Materials

COURSE OVERVIEW

Learning Outcomes

Part I

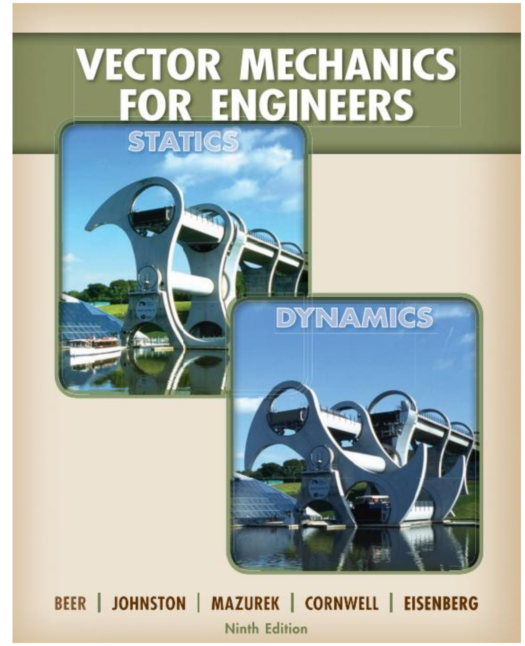
- Apply the concepts of *static equilibrium* of rigid body to solve for the unknown forces under *different supporting conditions* and *external loads*.
- Apply the concept of static equilibrium to analyze internal forces in *assembly of rigid bodies*, specially to *truss, frames and machines*.
- Describe the relationships between *internal force/moment* induced within *slender structure* and the corresponding stress states, which in turn define their structural integrity.

Part II

- Describe the *mechanical properties* of engineering materials and how they are tested.
- Correlated the structures of materials to their mechanical properties.
- Perform simple material selection.

Course information

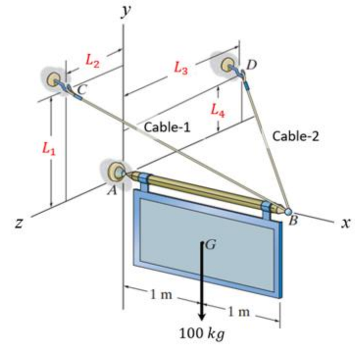
<table>
<tr><th>Week no.</th><th colspan="2">Topics to be covered</th></tr>
<tr><td>1</td><td colspan="2">Review on forces and moments</td></tr>
<tr><td>2</td><td colspan="2">Static of rigid body</td></tr>
<tr><td>3</td><td colspan="2">Static analysis of truss</td></tr>
<tr><td>4</td><td colspan="2">Static analysis of frames and machines</td></tr>
<tr><td>5</td><td colspan="2">Internal force/moment in slender structures</td></tr>
<tr><td>6</td><td>Principle of Virtual Works</td><td>More practice examples on static of rigid body and truss</td></tr>
<tr><td>7</td><td colspan="2">Mid-term quiz on static of rigid body & truss</td></tr>
<tr><td>7 to 13</td><td colspan="2">Part II by Prof Christina on Materials</td></tr>
<tr><td colspan="3">Final Exam on 1-Dec-2026 (Tuesday) 1:00 PM</td></tr>
</table>

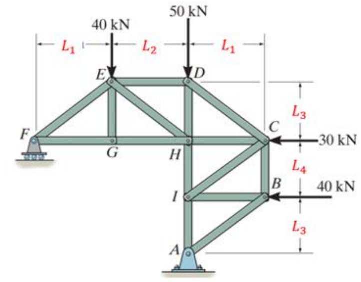


Course information…

- ***Experiments* (20 %)**
  - Lab 1 - Statics and Friction.
  - Lab 2 - Tensile test on Silicone rubber.
  - Group of 3~4 students, One report per group.

- ***Modeling and Simulation using SOLIDWORKS* (10 %)**
  - Conduct static analyses of rigid body and truss structure.
  - Take home assignment.



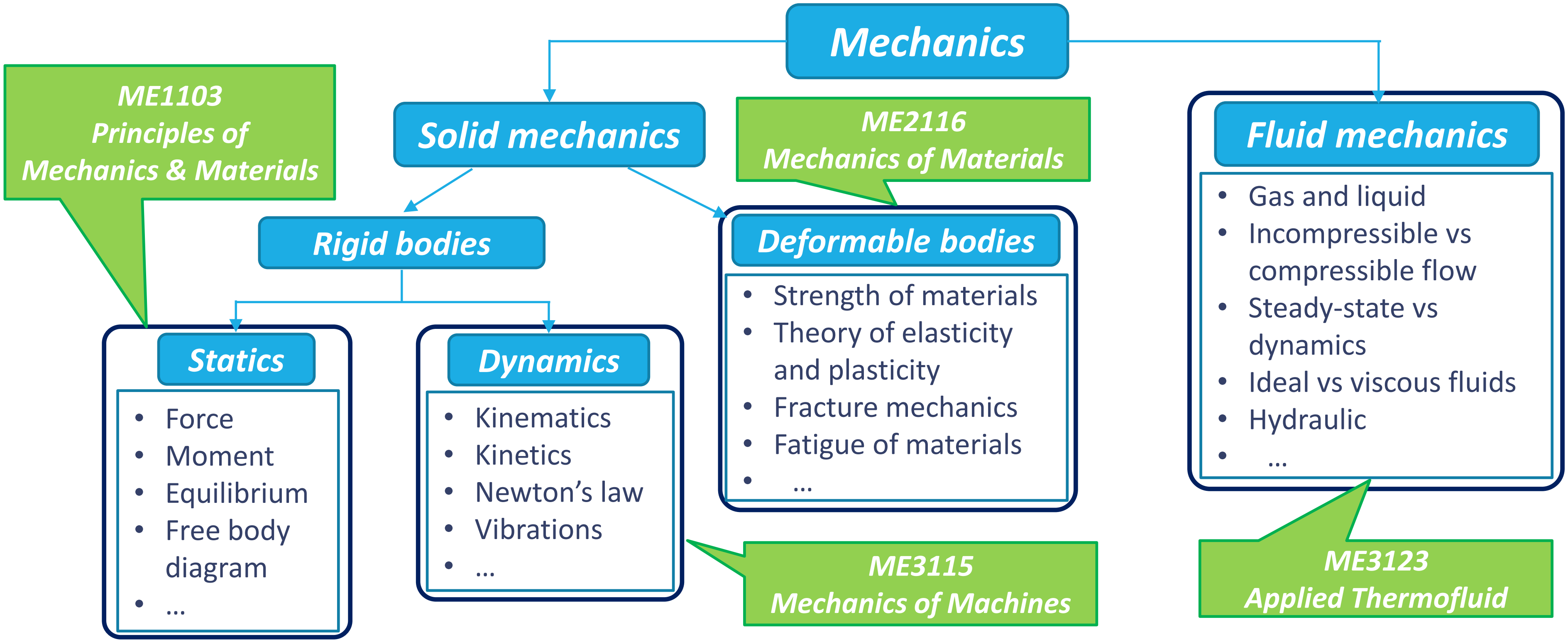


- ***One mid-term Quiz* (10 %)**
  - To be conducted physical in ***WEEK 7** (Tentatively on 29 Sep)*
  - Closed book assessment with *ONE-A4 cheat sheet*.

- ***Final Exam* (60 %)**
  - 2 hours (2 questions from Part I and short questions for Part II).
  - Closed book assessment with *ONE-A4 cheat sheet*.

What is Mechanics

- Mechanics is a branch of physical science which ***describes*** and/or ***predicts*** the conditions of ***rest*** or ***motion of bodies*** under ***the action of forces***.
- It is an applied science that uses ***extensive mathematics and physics*** to explain and predict physical phenomena and thus lay the foundation for engineering study.



Why Study Mechanics

- The main objective of strength of materials is to examine the load-carrying capacity of a body (system) from three standpoints:
  - Strength – can it withstand the applied loads?
  - Stiffness – does it deform excessively under the applied loads?
  - Stability – will it collapse under the applied loads?

Design for Failure!!!

- From engineering design perspective, this study help to establish the load limits of the body (system), and/or to achieve an optimize design (such as minimum weight, volume and cost).

- The basic principles of analysis consist of:
  - *Statics – Equilibrium of forces must be satisfied.*
  - Deformations – Applying stress-strain-displacement relations for material behavior to derive the deformations of body.
  - Geometry – Geometric fit or compatibility of deformations must be satisfied.